CONTINUOUS ASSESSMENT (CA)

Backend Development | Practical Assessment

Total Marks: 8	**Duration:** 1 Hour	**Technology:** Node.js, Express.js, MongoDB

Problem Statement: Library Book Management API

A college library wants to develop a backend API to manage its book records. Each book contains a title, author, category, availability status, and creation date. You are required to build a REST API using Node.js, Express.js, and MongoDB with Mongoose to create, retrieve, and update book records.

Subtasks and Marks

Subtask 1: Add a New Book — 3 Marks

Implement **POST /api/books** to add a book to MongoDB.
• Create a suitable Mongoose schema and model.
• Accept **title**, **author**, and **category** in the request body.
• Validate that the title is provided and return an appropriate response.

Subtask 2: Retrieve Books with Pagination — 3 Marks

Implement **GET /api/books** to retrieve books from MongoDB.
• Accept **page** and **limit** as query parameters.
• Use **skip()** and **limit()** to implement pagination.
• Return the books along with the current page and total number of records.

Subtask 3: Update Book Availability — 2 Marks

Implement **PATCH /api/books/:id** to update a book's availability status.
• Accept the new availability status in the request body.
• Update the record using its MongoDB ID.
• Return an appropriate error response if the book does not exist.

Evaluation Rubric

Criterion	Marks
Mongoose schema/model and book creation API	3
Pagination using query parameters, skip() and limit()	2
Pagination metadata and total record count	1
Correct update API using book ID	1
Appropriate HTTP status codes and error handling	1
Total	**8**

Student Instructions

• Use Node.js, Express.js, and MongoDB with Mongoose.

• Test endpoints using Postman, Thunder Client, or another REST client.

• Submit the source code and API response screenshots as instructed.

• Authentication, frontend development, and deployment are not required.

Suggested time plan: Setup and schema — 10 min | Create API — 15 min | Pagination — 20 min | Update API and testing — 15 min